Problem 1: Check the properties of the following rotation matrix

$$ {}^{A}_{B}R = \begin{bmatrix} 0.82364 & 0.1 & 0.5582 \\ 0.4755 & 0.41456 & -0.7759 \\ -0.30902 & 0.9045 & 0.2939 \end{bmatrix} $$

MAGNITUDE OF THE ROWS:

ROW 1: $(0.82364)^2 + (0.1)^2 + (0.5582)^2 = 1$

ROW 2: $(0.4755)^2 + (0.41456)^2 + (-0.7759)^2 = 1$

ROW 3: $(-0.30902)^2 + (0.9045)^2 + (0.2939)^2 = 1$

MAGNITUDE OF THE COLUMNS:

COLUMN 1: $(0.82364)^2 + (0.4755)^2 + (-0.30902)^2 = 1$

COLUMN 2: $(0.1)^2 + (0.41456)^2 - (0.9045)^2 = 1$

COLUMN 3: $(0.5582)^2 + (-0.7759)^2 + (0.2939)^2 = 1$

CROSS PRODUCTS OF ROWS & COLUMNS:

$$ \underbrace{\{0.82364,\ 0.1,\ 0.5582\}}_{\text{ROW 1}} \times \underbrace{\{0.4755,\ 0.4145,\ -0.7759\}}_{\text{ROW 2}} $$

$$ = \underbrace{\{-0.309,\ 0.9045,\ 0.2939\}}_{\text{ROW 3}} $$

ROW 2 × ROW 3 = ROW 1

ROW 3 × ROW 1 = ROW 2

THIS PROPERTY CAN BE CHECKED FOR COLUMNS

$$ \begin{Bmatrix} 0.1 \\ 0.41456 \\ 0.9045 \end{Bmatrix} \times \begin{Bmatrix} 0.5582 \\ -0.7759 \\ 0.2939 \end{Bmatrix} = \begin{Bmatrix} 0.82364 \\ 0.4755 \\ -0.30902 \end{Bmatrix} $$

COLUMN 2 × COLUMN 3 = COLUMN 1

COLUMN 1 × COLUMN 2 = COLUMN 3

COLUMN 3 × COLUMN 1 = COLUMN 2

DOT PRODUCTS OF ROWS & COLUMNS:

ROW 1 • ROW 2 $= (0.82364)(0.4755) + (0.1)(0.41456) + (0.5582)(-0.7759) = 0$

ROW 2 • ROW 3 $= (0.4755)(-0.30902) + (0.41456)(0.9045) + (-0.7759)(0.2939) = 0$

ROW 3 • ROW 1 $= (-0.30902)(0.82364) + (0.9045)(0.1) + (0.2939)(0.5582) = 0$

COLUMN 1 • COLUMN 2 $= (0.82364)(0.1) + (0.4755)(0.41456) + (-0.30902)(0.9045) = 0$

COLUMN 2 • COLUMN 3 $= (0.1)(0.5582) + (0.41456)(-0.7759) + (0.9045)(0.2939) = 0$

COLUMN 3 • COLUMN 1 $= (0.5582)(0.82364) + (-0.7759)(0.4755) + (0.2939)(-0.30902) = 0$

DETERMINANT OF R:

$$ |R| = 0.999997 \approx 1 $$

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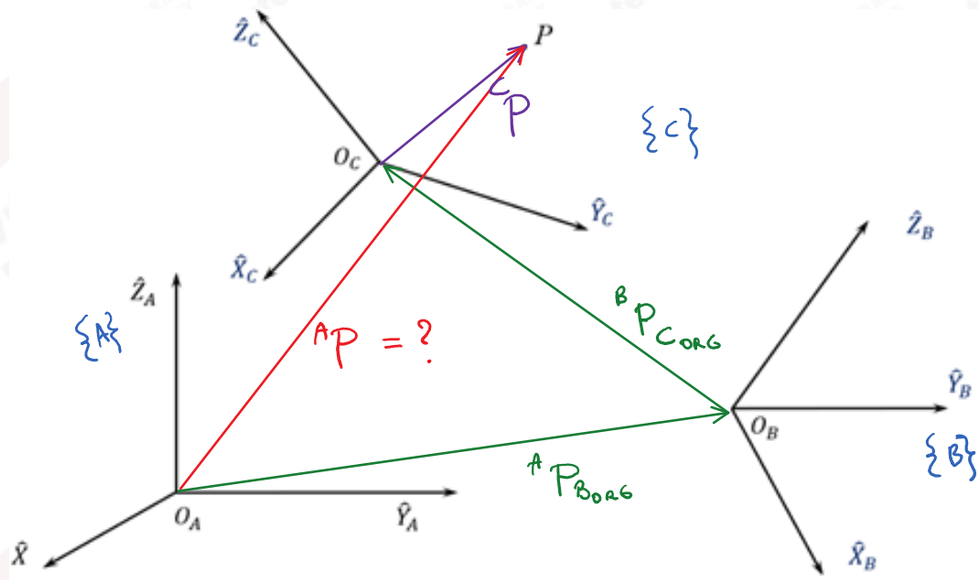
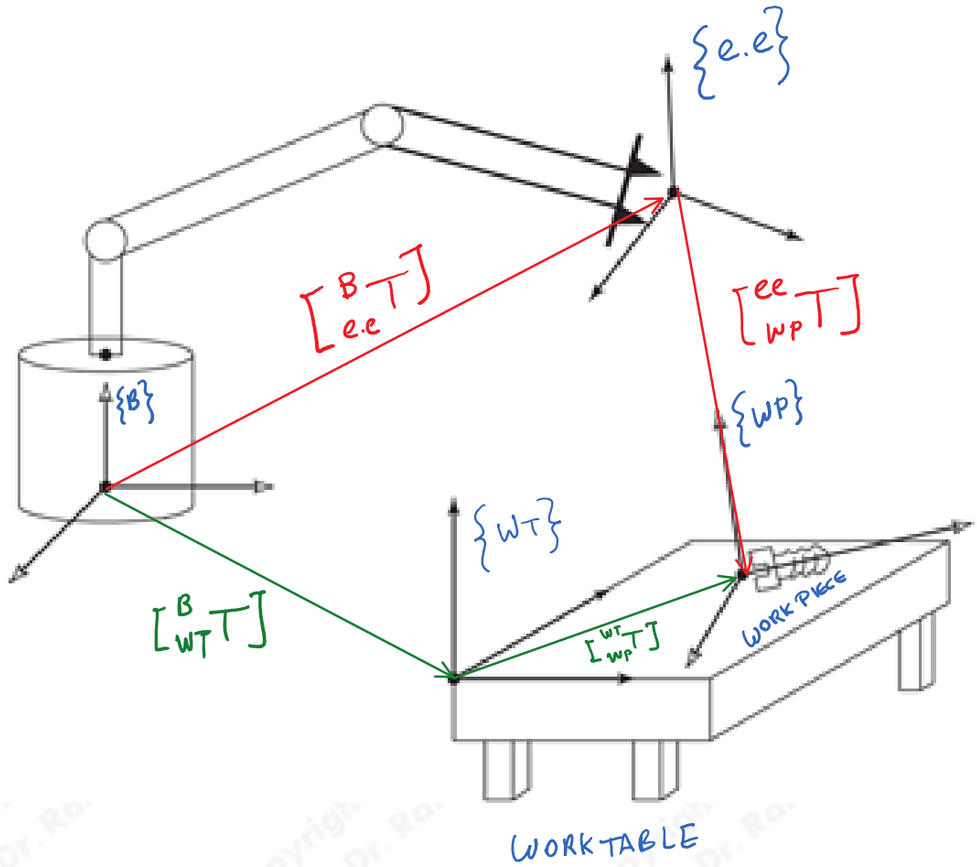
Problem 2: Determine ${}^{A}P_{\square}$, if the position vectors ${}^{A}P_{B_{ORG}}$, ${}^{B}P_{C_{ORG}}$, and ${}^{C}P_{\square}$ and the rotation matrices ${}^{A}_{B}R$ and ${}^{B}_{C}R$ are known.



$$
{}^{A}P = {}^{A}P_{B_{ORG}} + \underbrace{{}^{A}_{B}R\,{}^{B}P_{C_{ORG}}}_{{}^{A}P_{C_{ORG}}} + \underbrace{\overbrace{{}^{A}_{B}R\,{}^{B}_{C}R}^{{}^{A}_{C}R}\,\underbrace{{}^{C}P}_{{}^{B}P}}_{{}^{A}P}
$$

Problem 3: A manipulator requires to pick up a workpiece, determine ${}^{ee}_{wp}T$ based on the other homogeneous transforms.



THERE ARE TWO WAYS OF REACHING THE WORKPIECE:

VIA WORKTABLE AND VIA END-EFFECTOR

$$\left[{}^{B}_{e.e}T\right]\left[{}^{e.e}_{wp}T\right] = \left[{}^{B}_{WT}T\right]\left[{}^{WT}_{wp}T\right]$$

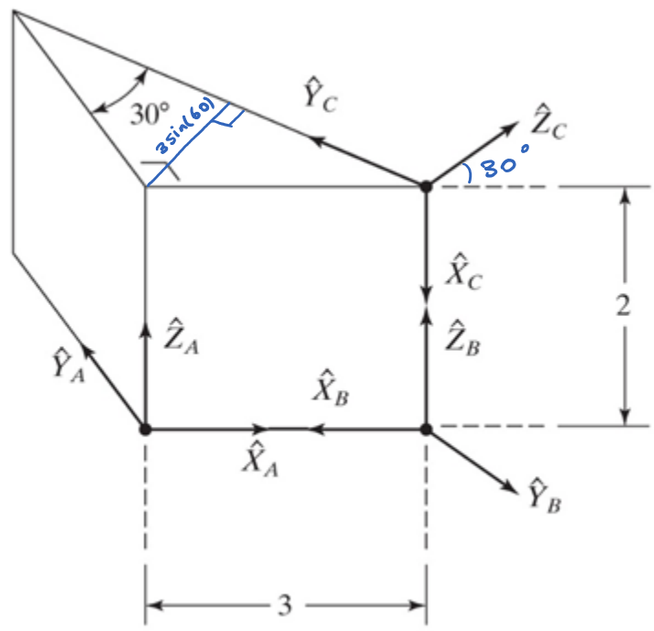
PRE-MULTPLY BOTH SIDES BY $\left[{}^{B}_{e.e}T\right]$:

$$\underbrace{\left[{}^{B}_{e.e}T\right]^{-1}\left[{}^{B}_{e.e}T\right]}_{I_3}\left[{}^{e.e}_{wp}T\right] = \left[{}^{B}_{e.e}T\right]^{-1}\left[{}^{B}_{WT}T\right]\left[{}^{WT}_{wp}T\right]$$

$$\left[{}^{e.e}_{wp}T\right] = \underbrace{\left[{}^{B}_{e.e}T\right]^{-1}}_{\left[{}^{B}_{e.e}T\right]^{T}}\left[{}^{B}_{WT}T\right]\left[{}^{WT}_{wp}T\right]$$

$$\left[{}^{e.e}_{wp}T\right] = \left[{}^{e.e}_{B}T\right]\left[{}^{B}_{WT}T\right]\left[{}^{WT}_{wp}T\right]$$

Problem 4: Determine the following homogeneous transforms of the frames attached to the corners of a wedge. ${}^{A}_{B}T$, ${}^{B}_{C}T$, ${}^{B}_{C}T$, *and* ${}^{C}_{A}T$.



$$[{}^{A}_{B}T] = \begin{bmatrix} [{}^{A}_{B}R]_{3\times3} & \{{}^{A}P_{B_{ORG}}\}_{3\times1} \\ 0 \quad 0 \quad 0 & 1 \end{bmatrix} = \begin{bmatrix} \{{}^{A}\hat{X}_B\}_{3\times1} & \{{}^{A}\hat{Y}_B\}_{3\times1} & \{{}^{A}\hat{Z}_B\}_{3\times1} & \{{}^{A}P_{B_{ORG}}\}_{3\times1} \\ 0 & 0 & 0 & 1 \end{bmatrix}_{4\times4}$$

$$= \begin{bmatrix} \begin{Bmatrix} -1 \\ 0 \\ 0 \end{Bmatrix} & \begin{Bmatrix} 0 \\ -1 \\ 0 \end{Bmatrix} & \begin{Bmatrix} 0 \\ 0 \\ +1 \end{Bmatrix} & \begin{Bmatrix} 3 \\ 0 \\ 0 \end{Bmatrix} \\ 0 & 0 & 0 & 1 \end{bmatrix}_{4\times4}$$

$$= \begin{bmatrix} -1 & 0 & 0 & 3 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & +1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}_{4\times4}$$

$$[{}^{B}_{C}T] = \begin{bmatrix} \{{}^{B}\hat{X}_C\} & \{{}^{B}\hat{Y}_C\} & \{{}^{B}\hat{Z}_C\} & \{{}^{B}P_{C_{ORG}}\} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & \cos(60) & -\cos(30) & 0 \\ 0 & -\sin(60) & -\sin(30) & 0 \\ -1 & 0 & 0 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$[{}^{A}_{C}T] = \begin{bmatrix} \{{}^{A}\hat{X}_C\} & \{{}^{A}\hat{Y}_C\} & \{{}^{A}\hat{Z}_C\} & \{{}^{A}P_{C_{ORG}}\} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -\cos(60) & \cos(30) & 3 \\ 0 & \sin(60) & \sin(30) & 0 \\ -1 & 0 & 0 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \equiv [{}^{A}_{B}T][{}^{B}_{C}T]$$

$$[{}^{C}_{A}T] = \begin{bmatrix} \{{}^{C}\hat{X}_A\} & \{{}^{C}\hat{Y}_A\} & \{{}^{C}\hat{Z}_A\} & \{{}^{C}P_{A_{ORG}}\} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & -1 & 2 \\ -\cos(60) & \cos(30) & 0 & 3\cos(60) \\ \cos(30) & \cos(60) & 0 & -3\sin(60) \\ 0 & 0 & 0 & 1 \end{bmatrix} \equiv [{}^{A}_{C}T]^{-1}$$

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Problem 5: First rotate $\{B\}$ about $\hat{X}_A$ by an angle $\gamma = 30^o$, then rotate $\{B\}$ about $\hat{Y}_A$ by an angle $\beta = -15^o$, and then rotate $\{B\}$ about $\hat{Z}_A$ by an angle $\alpha = 120^o$.

- Find the numerical rotation matrix.
- What would be the values of the angles if the $Z - Y - Z$ Euler angle representation was used?
- What would be the values of $\hat{K} = [k_x \quad k_y \quad k_z]$ and θ if equivalent angle-axis representation was used?

THIS IS THE FIXED ANGLE REPRESENTATION X-Y-Z (SLIDE 18)

USING SLIDE 19 :

$$ {}^A_B R_{XYZ}(\gamma, \beta, \alpha) = R_Z(\alpha)\, R_Y(\beta)\, R_X(\gamma) = \begin{bmatrix} c\alpha c\beta & c\alpha s\beta s\gamma - s\alpha c\gamma & c\alpha s\beta c\gamma + s\alpha s\gamma \\ s\alpha c\beta & s\alpha s\beta s\gamma + c\alpha c\gamma & s\alpha s\beta c\gamma - c\alpha s\gamma \\ -s\beta & c\beta s\gamma & c\beta c\gamma \end{bmatrix} $$

$$ = \begin{bmatrix} c(120)c(-15) & c(120)s(-15)s(30) - s(120)c(30) & c(120)s(-15)c(30) + s(120)s(30) \\ s(120)c(-15) & s(120)s(-15)s(30) + c(120)c(30) & s(120)s(-15)c(30) - c(120)s(30) \\ -s(-15) & c(-15)s(30) & c(-15)c(30) \end{bmatrix} $$

$$ = \begin{bmatrix} -0.483 & -0.6853 & 0.5451 \\ 0.8363 & -0.5451 & 0.0559 \\ 0.2588 & 0.483 & 0.8365 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} $$

Z-Y-Z EULER ANGLES: (FROM SLIDE 23)

$$ \beta = \mathrm{Atan2}\left(\sqrt{r_{31}^2 + r_{32}^2}, r_{33}\right) = \mathrm{Atan2}(0.548, 0.8365) = 33.23^\circ $$

$$ \alpha = \mathrm{Atan2}(r_{23}/s\beta, r_{13}/s\beta) = \mathrm{Atan2}(0.102, 0.9948) = 5.86^\circ $$

$$ \gamma = \mathrm{Atan2}(r_{32}/s\beta, -r_{31}/s\beta) = \mathrm{Atan2}(0.8814, -0.4723) = 118.18^\circ $$

EQUIVALENT ANGLE - AXIS REPRESENTATION

$$ \theta = \mathrm{acos}\left(\frac{r_{11} + r_{22} + r_{33} - 1}{2}\right) = \mathrm{acos}\left(\frac{-0.483 - 0.5451 + 0.8363 - 1}{2}\right) = 126.57^\circ $$

$$ \hat{K} = \frac{1}{2\sin(\theta)} \begin{bmatrix} r_{32} - r_{23} \\ r_{13} - r_{31} \\ r_{21} - r_{12} \end{bmatrix} = \frac{1}{2\sin(126.57^\circ)} \begin{Bmatrix} 0.4271 \\ 0.2588 \\ 1.5218 \end{Bmatrix} = \begin{Bmatrix} 0.2659 \\ 0.1782 \\ 0.9474 \end{Bmatrix} $$